

FDS: Pokemon Battles Prediction 2025

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1. Feature Engineering & Selection

The model's performance is critically dependent on the features extracted from the battle data. We engineered several features to capture the state of the battle after 30 turns.

1.1. Engineered Features

The following features were created to provide a quantitative snapshot of the battle's dynamics:

- **Mean HP Percentage** ('p1_mean_pc_hp', 'p2_mean_pc_hp'): The average remaining health across each player's team, indicating overall survivability.
- **Surviving Pokémon** ('p1_surviving_pokemon', 'p2_surviving_pokemon'): A count of Pokémon with HP greater than zero for each player, representing remaining strategic resources.
- **Status and Effects Score** ('p1_status_score', 'p2_status_score'): A composite score quantifying the impact of status conditions and other persistent effects on each team.
- **Base Stat Differentials**: To capture the intrinsic statistical advantages of the teams, we calculated the difference between the sum of base stats for Player 1's team and Player 2's known Pokémon. This was done for all six stats (HP, Attack, Defense, Special Attack, Special Defense, Speed).
- **Pokémon Changes** ('p1_n_changes', 'p2_n_changes'): A count of how many times each player switched their active Pokémon, which can be an indicator of strategic depth.

We also tried to implement different categorical features, like: type of Pokemon, name of Pokemon or Pokemon move type. None of those was able to boost our accuracy. We've tried a lot of features and feature combinations, but the ones present are what worked best in the end.

1.2. Feature Selection

The final features were selected based on their intuitive relevance to a battle's outcome. This set provides a balanced view of the battle state, capturing team health, strategic actions, and raw statistical power without being overly complex.

2. Model Experimentation and Refinement

2.1. Baseline Model: Logistic Regression

A Logistic Regression model [2] was chosen as the baseline for this classification task. It is a linear model well-suited for binary classification problems due to its interpretability, computational efficiency, and strong performance on linearly separable data.

2.2. Data Scaling and Refinement

Features were scaled using 'StandardScaler' to ensure that each feature contributed equally to the model's training process. While Logistic Regression provided a solid baseline, more complex models like Gradient Boosting Machines [1] could be explored to capture non-linear relationships and improve performance.

3. Validation

A robust validation strategy is critical for assessing the model's ability to generalize to unseen data.

3.1. Stratified K-Fold Cross-Validation

We employed a 5-fold Stratified K-Fold cross-validation [3] strategy. This involves splitting the training data into 5 folds, training the model on 4 folds, and validating on the 5th. This process is repeated 5 times, with each fold serving as the validation set once. Stratification ensures that the class distribution (win/loss) is consistent across all folds, preventing biased evaluation.

3.2. Evaluation Metrics

Model performance was assessed using a suite of metrics:

- **Accuracy**: The primary metric, representing the proportion of correct predictions.
- **Precision, Recall, and F1-Score**: These metrics provide a more nuanced view of the model's performance, especially in cases of class imbalance.

Averaging these metrics across the 5 folds gives a reliable estimate of the model's true performance. The final test predictions were generated by averaging the outputs of the 5 models trained during cross-validation, an ensembling technique that enhances prediction stability.

References

- [1] Zhiyuan He, Danchen Lin, Thomas Lau, and Mike Wu. Gradient boosting machine: A survey, 2019. [1](#)
- [2] Michael P LaValley. Logistic regression. *Circulation*, 117(18):2395–2399, 2008. [1](#)
- [3] Slamet Widodo, Herlambang Brawijaya, and Samudi Samudi. Stratified k-fold cross validation optimization on machine learning for prediction. *Sinkron*, 7:2407–2414, 2022. [1](#)